EXPERIMENT - 2

Aim :- Convert the ER diagram into a relational Schema : Write complete MYSQL CREATE TABLE statements which primary Keys, foreign Keys, NOT NULL, UNIQUE, ON DELETE CASCADE and SET NULL constraints. Insert sample data and demonstrate referential integrity violations.

Source Code :-

```
DROP DATABASE IF EXISTS e-commerce;
CREATE DATABASE e-commerce;
USE e-commerce;
--1. CUSTOMER TABLE
CREATE TABLE Customer (
    Customer_id INT PRIMARY KEY,
    Name VARCHAR (50) NOT NULL,
    Email VARCHAR (100) UNIQUE,
    mob_num VARCHAR (15)
);
--2. ADDRESS TABLE
CREATE TABLE Address (
    Customer_id INT,
    address_type VARCHAR (20),
    PRIMARY KEY (Customer_id, address_type),
    FOREIGN KEY (customer_id)
        REFRENCES Customer (customer_id)
);
--3. SELLER TABLE
CREATE TABLE SELLER (
    Sum_id INT PRIMARY KEY
);
```

```
--4. CATEGORY TABLE
CREATE TABLE Category (
    Category_id INT PRIMARY KEY,
    Category_name VARCHAR (50) NOT NULL,
    discription VARCHAR (200)
);
--5. PRODUCT TABLE
CREATE TABLE Product (
    Product_id INT PRIMARY KEY,
    Product_name VARCHAR (100) NOT NULL,
    discription VARCHAR (200),
    price DECIMAL (10,2) NOT NULL,
    Seller_id INT,

    FOREIGN KEY (Seller_id)
      REFRENCES Seller (Seller_id)
);
--6. ELECTRONICS TABLE
-- Specialization of product
CREATE TABLE Electronics (
    product_id INT PRIMARY KEY,
    Brand VARCHAR (50),
    Warranty VARCHAR (50),
    FOREIGN KEY (product_id)
      REFRENCES Product (product_id)
);
-7. FASHION TABLE
CREATE TABLE Fashion (
    product_id INT PRIMARY KEY,
    Size VARCHAR (20),
    material VARCHAR (50),
    FOREIGN KEY (product_id)
      REFRENCES Product (product_id)
);
```

```
--8. HOME APPLIANCES TABLE
CREATE TABLE Home_Appliances (
    product_id INT PRIMARY KEY,
    power_rating VARCHAR (50),
    capacity VARCHAR (50),
    FOREIGN KEY (product_id)
      REFRENCES Product (product_id)
);
--9. ORDERS TABLE
CREATE TABLE Orders (
    order_id INT PRIMARY KEY,
    order_data DATE,
    total_amount DECIMAL (10,2),
    discount DECIMAL (10,2),
    Customer_id INT,
    FOREIGN KEY (customer_id)
      REFRENCES Customer (customer_id)
);
--10. ORDER_ITEM TABLE
-- Weak Entity
-- Composite Primary Key
CREATE TABLE Order_Item (
    order_id INT,
    product_id INT,
    quantity INT NOT NULL,
    discount DECIMAL (10,2),
    Category_id INT,
    PRIMARY KEY (order_id, product_id)
    FOREIGN KEY (order_id)
      REFRENCES Orders (order_id),
    FOREIGN KEY (product_id)
      REFRENCE product (product_id)
);
```

```
--11. PAYMENT TABLE
CREATE TABLE Payment (
    payment_id INT PRIMARY KEY,
    Amount DECIMAL (10,2) NOT NULL,
    Customer_id INT,
    Seller_id INT,
    FOREIGN KEY (customer_id)
      REFRENCES Customer (customer_id),
    FOREIGN KEY (Seller_id)
      REFRENCES Seller (Seller_id)
);
--12. DELIVERY TABLE
CREATE TABLE DELIVERY (
    delivery_id INT PRIMARY KEY,
    tracking_no VARCHAR (50),
    delivery_date DATE,
    estimated_date DATE,
    product_id INT,
    FOREIGN KEY (product_id)
      REFRENCES Product (product_id)
);
-- INSERT DATA
INSERT INTO Customer VALUES
(101, 'Parul', 'parul@gmail.com', 9874540312),
(102, 'Aanya', 'aanya@gmail.com', 7943568019),
SELECT * FROM Customer;
-- ADDRESS DATA
INSERT INTO Address VALUES
(101, 'HOME'), (101, 'Work'), (102, 'Home').
SELECT * FROM Address;
--SELLER DATA
INSERT INTO Seller Values (201), (202)
SELECT * FROM Seller;
```

```
-- Category Data
INSERT INTO Category VALUES
(301, 'Electronics', 'Electronics products'),
(302, 'Fashion', 'Fashion products'),
(303, 'Home Appliance', 'Home appliance');
SELECT * FROM Category;
-- PRODUCT DATA
INSERT INTO Product VALUES
(401, 'Laptop', 'DellLaptop', 55000.00, 201),
(402, 'T-Shirt', 'Cotton T-shirt', 999.00, 202),
(403, 'Washing Machine', 'Automatic Washing Machine', 30000.00, 201);
SELECT * FROM Products;
-- ELECTRONICS DATA
INSERT INTO Electronics VALUES
(401, 'Dell', '2 Years');
SELECT * FROM Electronics;
-- FASHION DATA
INSERT INTO Fashion VALUES
(402, 'M', 'Cotton');
SELECT * FROM Fashion;
-- HOME APPLIANCES DATA
INSERT INTO Home_Appliances VALUES
(403, '2000W', '7Kg');
SELECT * FROM Home_Appliances;
-- ORDER DATA
INSERT INTO Order_Item VALUES
(501, 401, 1, 500.00, 301),
(501, 402, 1, 0.00, 302),
(502, 403, 1, 1000.00, 303);
SELECT * FROM Order_Item;
```

```
--PAYMENT DATA
INSERT INTO Payment VALUES
(601, 55499.00 , 101, 201),
(602, 29000.00 , 102, 201);
SELECT * FROM Payment;

-- DELIVERY DATA
INSERT INTO Delivery VALUES
(701, 'TRK1001', '2026-08-24', NULL, 401),
(702, 'TRK1002', '2026-08-25', NULL, 403);
SELECT * FROM Delivery;
```

OUTPUT :- Customer table

1.

Customer_id	Name	Email	mob_num
101	Parul	parul@gmail.com	9221879410
102	Aanya	aanya@gmail.com	9238478292
NULL	NULL	NULL	NULL

2. Product table

product_id	product_name	description	price	Seller_id
401	Laptop	Dell Laptop	55000.00	201
402	T-shirt	cotton T-shirt	999.00	202
403	Washing Machine	Automatic Washing machine	30000.00	201
NULL	NULL	NULL	NULL	NULL

3. Order_Item table

Order_id	product_id	quantity	discount	category_id
501	401	1	500.00	301
502	402	1	0.00	302
502	403	1	1000.00	303
NULL	NULL	NULL	NULL	NULL

4. Delivery Table

deliver_id	tracking_no	estimated	delivery_date	product_id
701	TRK1001	2026-08-24	NULL	401
702	TRK1002	2026-08-25	NULL	403
NULL	NULL	NULL	NULL	NULL